A BOUND ON THE DEGREES OF THE ELLIPTIC DECOMPOSITION OF A FUNCTION WITH N POLES

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Editorial note. Assembly history: r001 failed with 'Undefined control sequence' at \prodtheta because this source declares its math macros (lines 22-27) AFTER \begin{document}, so a preamble span that stops at \begin{document} misses them. r002 adds that run as a second preamble span (pure command definitions, copied verbatim); the generator now detects such post-document macro runs automatically. No mathematics changed.

Allowed standard prerequisites (editorial).

- The identification of functions periodic with respect to k Lambda with Lambda-periodic functions by scaling, and the even/odd decomposition formulas for f in terms of A(weierstrass) and weierstrass' times B. Used inside the counted proof at lines 627-630 and 631-634, 643-646. Reduces to $k = 1$ and produces the explicit expressions whose pole counts are estimated.

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- Elliptic functions have as many zeroes as poles, counted with multiplicity. Used inside the counted proof at lines 640-641. Turns the pole bound on A(weierstrass) into the degree bound on its numerator.

Lemma 0.1. *Assume that $f \in \mathrm{M}_{k\Lambda}$, seen has a meromorphic function over $\mathbb{C}/k\Lambda$, has at most N poles counted with multiplicities (or, equivalently, that $f = p/q$ with $p, q \in \Theta_k$ such that $\deg_k p, \deg_k q \leq N$). Then, there exist $A = P/Q$ and $B = R/S$ with $P, Q \in \mathbb{C}[X]$ of degree at most $2N$ and $R, S \in \mathbb{C}[X]$ of degree at most $2N + 3$ such that*

$$f = A(\wp_k) + \wp_k' B(\wp_k).$$

Proof. Using the fact that $f(z)$ belongs to $\mathrm{M}_{k\Lambda}$ if and only if $f(kz)$ belongs to M_Λ, it is easily seen that it is sufficient to prove the lemma for $k = 1$. In what follows, we see the Λ-periodic elliptic functions as meromorphic functions on $\mathbb{C}/\Lambda$. Let $A, B \in \mathbb{C}(X)$ be such that $f = A(\wp) + \wp' B(\wp)$. It follows from the formula

$$A(\wp(z)) = \frac{f(z) + f(-z)}{2}$$

that $A(\wp)$ has at most $2N$ poles counted with multiplicities in $\mathbb{C}/\Lambda$. But, if $A = P/Q$ with $\gcd(P, Q) = 1$ then $A(\wp)$ has at least $\deg Q$ poles counted with multiplicities in $\mathbb{C}/\Lambda$ (namely, the zeroes of $Q(\wp)$). So $\deg Q \leq 2N$.

Using the fact that elliptic functions have the same numbers of zeroes and poles, the same argument applied to $1/A(\wp)$ shows that $\deg P \leq 2N$.

Using the formula

$$B(\wp(z)) = \frac{f(z) - f(-z)}{2\wp'(z)},$$

similar arguments show that $\deg R \leq 2N + 3$ and $\deg S \leq 2N + 3$. □